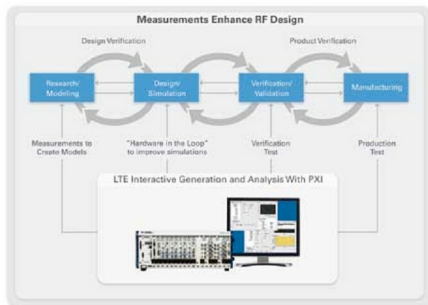




Next-generation wireless standards testing with PXI

antennae on a single mobile device, next-generation standards come with substantially more difficult hardware requirements. As a result, the measurement and instrumentation required for next-generation wireless standards are more challenging as well."

Equipment for in-line processing of RF signals. Another evolution is happening in the in-line processing of RF signals. Samanta explains, "With bandwidth increasing, integrating processors like field-programmable gate arrays (FPGAs) and digital signal processors (DSPs) with RF modules like vector signal analysers (VSAs) and vector signal generators (VSGs) allows users to react to the device under test (DUT) in real time and make changes accordingly."

Equipment to generate custom-built signals. Another key challenge before RF designers is to generate custom-built signals (for evolving wireless standards and defence applications) to test their systems without sharing the intellectual property. "Wideband arbitrary waveform generators with

RF bandwidths up to 5 GHz and microwave vector signal generators up to 44 GHz provide such solutions. Additionally, new signal analysers with up to 900MHz bandwidth and millimetre-wave oscilloscopes with up to 33 GHz of real-time bandwidth overcome the challenge to analyse these signals," explains Vishal.

Advanced test equipment for strategic electronics. Strategic electronics systems like radars with better speed and range resolutions are pushing the need for millimetre-wave test equipment touching terahertz-range.

Vishal affirms, "Wide-band signal analysers with up to 900MHz intermediate frequency (IF) bandwidth and 160MHz analysis bandwidth have been introduced. The same is the requirement for testing high-bandwidth future radars and evolving wireless standard 802.11ac. The combination of a wide-bandwidth ARB with a microwave vector signal generator provides flexibility to generate custom waveforms and signals of up to 2GHz bandwidth, 44GHz carrier frequency and waveform for virtually any type of wireless standard and electronic warfare signal simulation."

Oscilloscopes for complex signal analysis. Conventionally, RF equipment are meant to perform analysis in frequency domain. But there has always been a need to capture and analyse signals at these high frequencies in time domain as well. Introduction of 33GHz millimetre-wave oscilloscopes has enabled RF engineers to capture and analyse these signals in time domain. Along with a versatile analysis tool like VSA, these oscilloscopes make the most convenient tools for complex signal analysis.

Advanced vector network analysers (VNAs). As the demand for faster data transfer rates and wider channel capacities continues to rise, precision RF components are essential to provide high-quality signals required

from base-band to RF. For their exceptional accuracy and flexibility, VNAs have become an indispensable tool on the design bench as well as in validation test systems on the production line.

Samant underlines the importance of using VNAs: "The unique hardware architecture of VNA allows to precisely characterise the DUT. VNA allows to understand the electrical contribution of each component, whether in



RF vector network analyser